Testing Walks Along with SQA

SQA stands for **Software Quality Assurance**. It's a critical field within software development focused on ensuring that software products meet certain standards of quality before being released to users.

Here’s a breakdown of what SQA involves:

1. **Definition**

SQA is a **systematic process** that ensures the quality of software throughout the **entire development lifecycle**. It includes activities like:

* Requirements analysis

* Test planning and execution

* Code reviews

* Process monitoring

* Defect tracking

* Quality audits

2. **Main Goals**

* Prevent defects rather than just find them
* Ensure software meets both **functional** and **non-functional** requirements
* Improve development processes to produce better quality products

3. **Key Activities**

* **Requirement Review** – making sure requirements are clear, complete, and testable.
* **Test Planning** – defining scope, approach, resources, and schedule for testing.
* **Test Case Design** – creating step-by-step procedures to verify that the application works as expected.
* **Manual Testing** – performing tests without automation tools.
* **Automation Testing** – using tools like Selenium, Playwright, etc., to automate test execution.
* **Bug Tracking and Reporting** – using tools like JIRA to log, prioritize, and manage bugs.
* **Performance & Security Testing** – verifying speed, stability, and safety of applications.

4. **Common Tools**

* **Test Management**: TestRail, Zephyr, Qase
* **Bug Tracking**: JIRA, Bugzilla
* **Automation**: Selenium, Playwright, Cypress
* **CI/CD Integration**: Jenkins, GitHub Actions
* **API Testing**: Postman, SoapUI, JMeter

5. **Roles in SQA**

* **SQA Engineer** – ensures process adherence and quality practices.

* **QA Analyst/Tester** – performs test planning, execution, and reporting.

* **Automation Engineer** – develops scripts to test software automatically.

* **Test Lead/Manager** – plans testing strategy and coordinates the QA team.

6. **Importance of SQA**

* Reduces cost by catching bugs early

* Improves customer satisfaction

* Enhances product reliability and performance

* Helps maintain compliance with standards (like ISO, CMMI)

7. **Difference Between QA & Testing**

QA (Quality Assurance)	Testing
Process-oriented	Product-oriented
Prevents defects	Detects defects
Proactive	Reactive
Focuses on process improvement	Focuses on identifying bugs